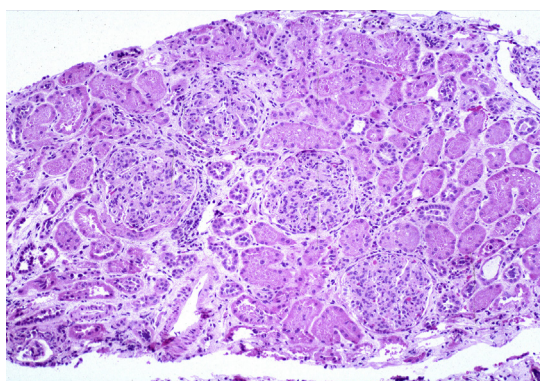


Postinfectious Glomerulonephritis

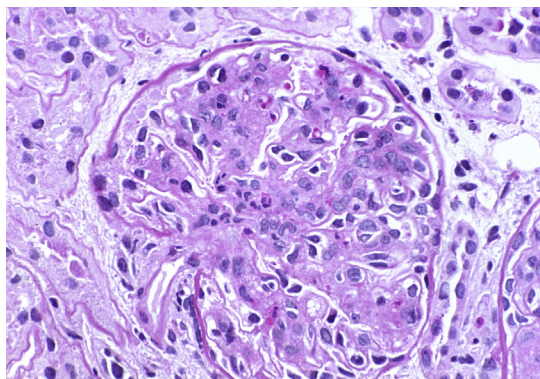
Agnes Fogo, MD

The AJKD Atlas of Renal Pathology presents a compilation of figures on a specific pathologic entity. You may download the figures to create your own personal, non-commercial library of images or to create slides for teaching purposes.



Copyright © 1998 by the National Kidney Foundation

Fig 1. Glomeruli show diffuse hypercellularity due to mesangial and endothelial cell increase and a large number of polymorphonuclear neutrophils (PMNs). These light microscopic findings are typical of diffuse proliferative acute postinfectious glomerulonephritis (hematoxylin & eosin; original magnification $\times 100$).



Copyright © 1998 by the National Kidney Foundation

Fig 2. Diffuse proliferative acute postinfectious glomerulonephritis with numerous PMNs with PAS-positive cytoplasm and endocapillary proliferation (periodic acid-Schiff; original magnification $\times 400$).

From the Department of Pathology, Vanderbilt University Medical Center, Nashville, TN.

Medical Photographer: Brent Weedman; with assistance from Kim Solez, MD, Joseph DeAlmeida, and Brenda Kalynchuk, of the National Kidney Foundation's cyberNephrology™ Team.

Address author queries to Agnes Fogo, MD, Department of Pathology, Vanderbilt University Medical Center, MCN C-3310, Nashville, TN 37232. E-mail: Agnes.Fogo@vanderbilt.edu

Am J Kidney Dis. 31(5):E1-E2.2.

© 1998 by the National Kidney Foundation, Inc. Published by Elsevier Inc. All rights reserved.

0272-6386/\$36.00

[http://dx.doi.org/10.1053/S0272-6386\(13\)90004-4](http://dx.doi.org/10.1053/S0272-6386(13)90004-4)

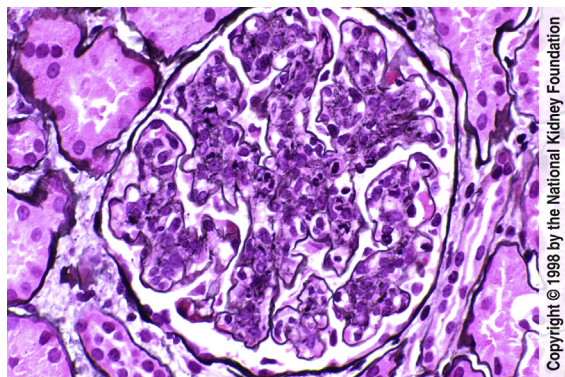


Fig 3. The large number of PMNs typical of acute poststreptococcal glomerulonephritis is evident in this high-power silver stain. The glomerular basement membrane does not show splitting or spikes. There is endocapillary proliferation, ie, proliferation of endothelial and mesangial cells and infiltrating cells in addition to filling and distending capillary loops. The large number of PMNs is characteristic of postinfectious glomerulonephritis, with the prototype being poststreptococcal glomerulonephritis (Jones' silver stain; original magnification $\times 400$).

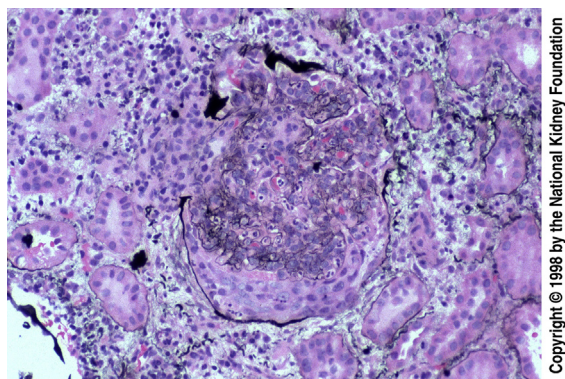


Fig 4. Crescents may occasionally be found in diffuse proliferative acute postinfectious glomerulonephritis, particularly in those patients who undergo renal biopsy when spontaneous resolution does not occur. The crescent in this case has ruptured Bowman's capsule, and there is an intense lymphoplasmocytic interstitial infiltrate surrounding the glomerulus (Jones' silver stain; original magnification $\times 200$).

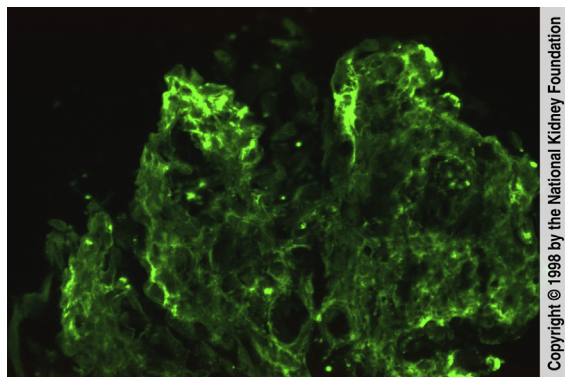


Fig 5a. Acute postinfectious glomerulonephritis by immunofluorescence typically shows irregular capillary loop deposits staining for IgG, seen here, or C3, seen in Figure 6 (immunofluorescence with anti-IgG; original magnification $\times 400$).

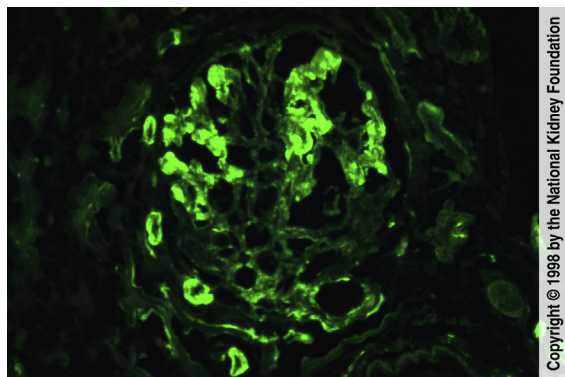


Fig 5b. The garland pattern of immune complexes due to large subepithelial deposits in acute postinfectious glomerulonephritis is shown (immunofluorescence with anti-IgG; original magnification $\times 400$).

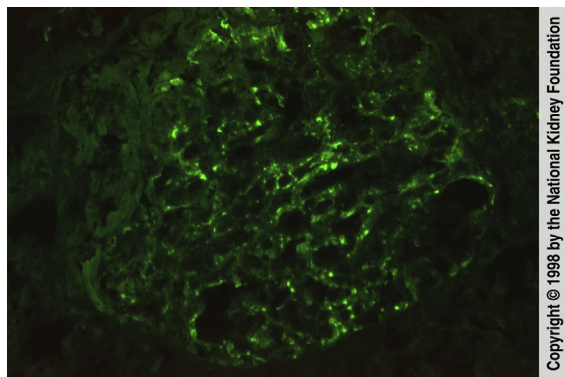


Fig 6. The scattered capillary wall granular deposits in acute poststreptococcal glomerulonephritis also stain for complement (immunofluorescence with antibody to C3; original magnification $\times 400$).

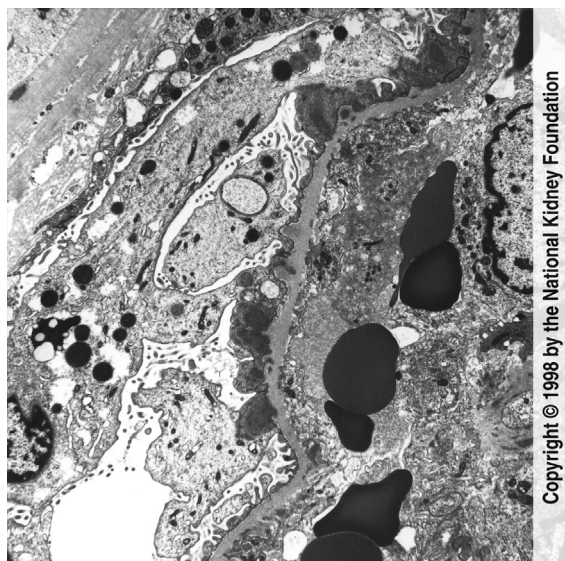
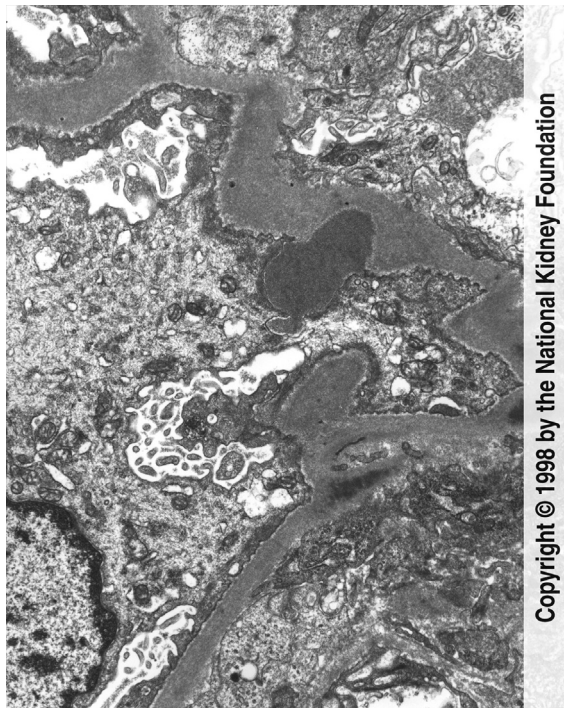


Fig 7. Large, subepithelial, hump-like scattered deposits in acute postinfectious glomerulonephritis. The deposits are irregularly spaced and frequently mottled, and they sit on top of the basement membrane with little, if any, basement membrane reaction (transmission electron micrograph; original magnification $\times 6,800$).



Fig 8. Hump- or dome-shaped deposits in acute postinfectious glomerulonephritis with extensive foot process effacement and endocapillary proliferation (transmission electron micrograph; original magnification $\times 11,250$).



Copyright © 1998 by the National Kidney Foundation

Fig 9. In this case of acute postinfectious glomerulonephritis, only rare subepithelial deposits remain with occasional intramembranous deposits. At this later stage, the deposits may be found only in the notch area, ie, where the basement membrane reflects over the mesangium. Such deposits may indent the basement membrane as well as protrude above it, as shown (transmission electron micrograph; original magnification, $\times 17,125$).